

Proactive Device Upgrade

A Machine Learning–Driven Retention Strategy
for High-Value Telecom Customers

· Telecommunications · Colombia Market

Dataset: 100,000 customers · XGBoost Classifier · Segment: High-Value + Aged Device

MARKET CONTEXT — Colombia Telecom

USD 7.4B

Market size
(2024)

ICEX / Expert Market Research, 2025

94%

Market share:
Claro + Tigo + Movistar

Deloitte Colombia, 2023

5–25×

Cost to acquire
vs. retain a customer

GoContact, 2024

Colombia's telecom market is dominated by 3 operators in a near-saturated, competitive landscape. Growth comes from retention and ARPU lift, not new subscribers.

In Q1 2025, number portability represented only 15.8% of total line withdrawals (CRC, 2025) — the remaining 84.2% is silent churn: disengagement without switching. This is where a predictive model adds value.

ARPU is under pressure: prepaid top-ups fell 5–7% YoY in early 2025 amid 5% inflation and 2.5% GDP growth. Retaining high-value customers is the priority lever (Bancolombia, 2026).

EDA — Device Ageing: The Strongest Churn Predictor

Primary finding: Device ageing emerged as the strongest predictor of customer churn in the 100,000-customer dataset.

421 days

Average device age (churners)

↓ Difference: +58 Days ↓

363 Days

Average device age (non-churners)



Device Age as the Core Driver

The variable eqpdays represents the age of the user's current mobile device.

Churners hold devices that are **58 days older** on average compared to active loyal customers.

This variable was classified as the **#1 feature by XGBoost gain**, proving that tech obsolescence triggers active competition search.

EDA — Key Findings from 100,000 Customers



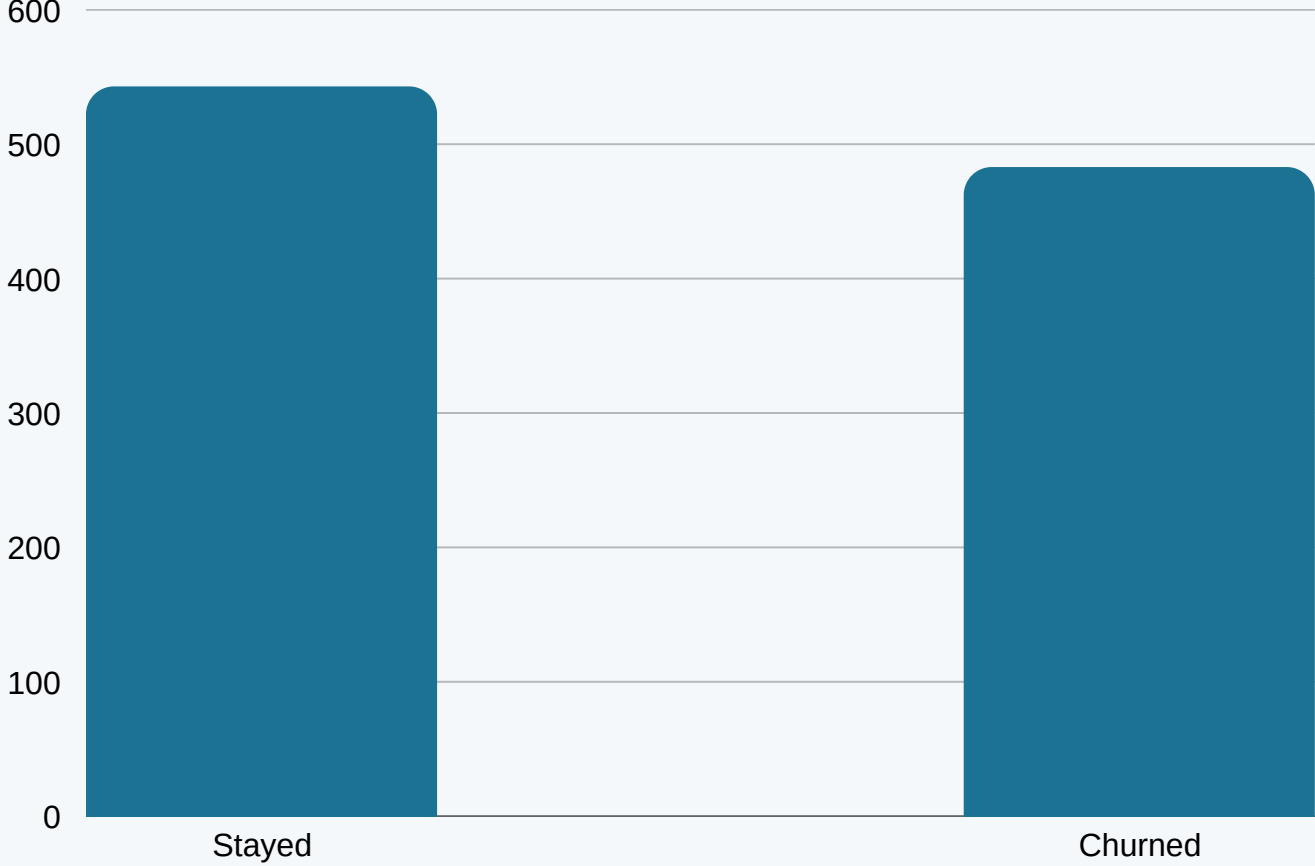
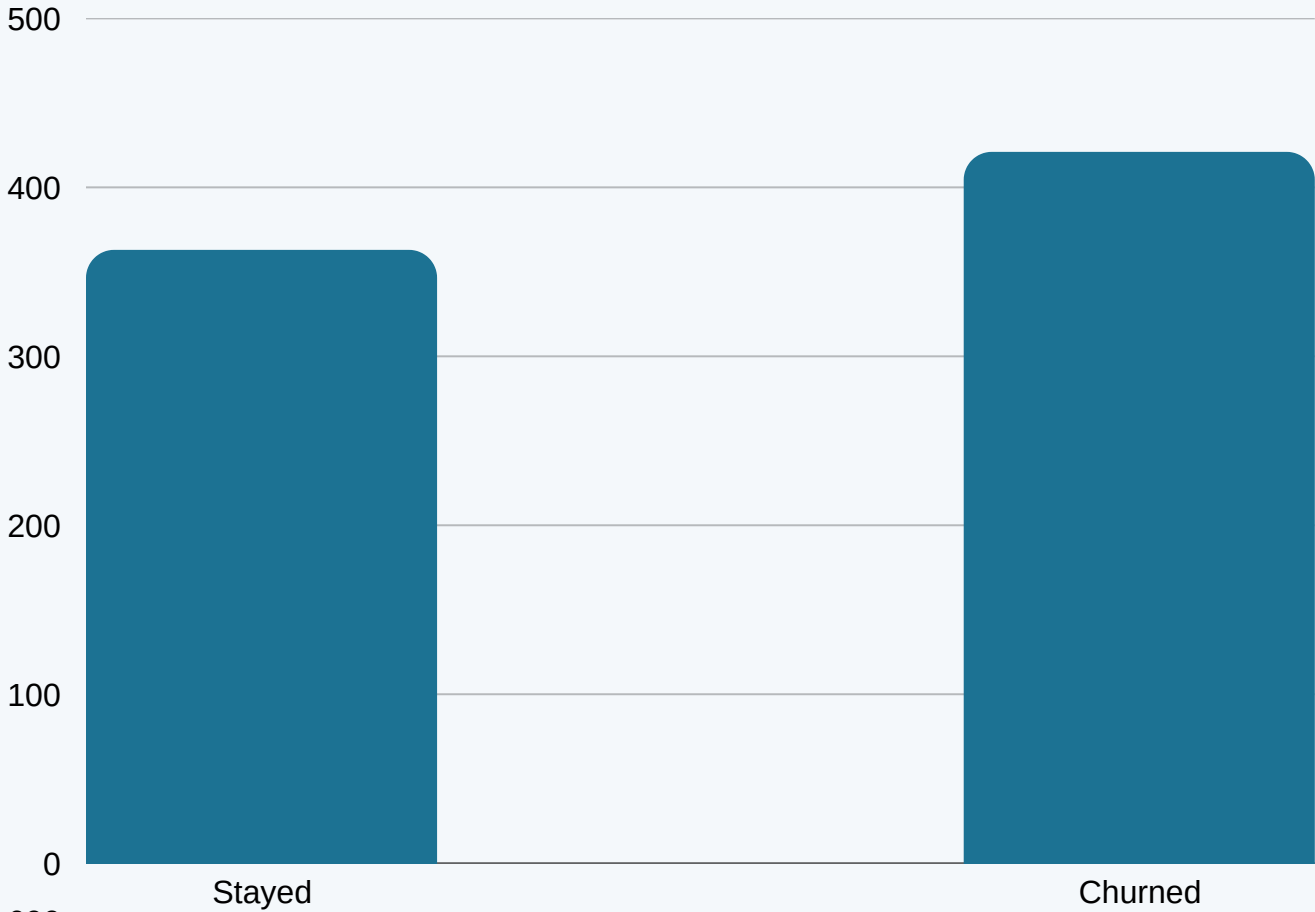
Balanced target

Global churn rate: 49.6% — classes are balanced, making accuracy a valid primary metric.



Device age is the strongest signal

Churners have devices 58 days older on average (421 vs 363 days). eqpdays is the #1 feature by XGBoost gain.



EDA — Key Findings from 100,000 Customers



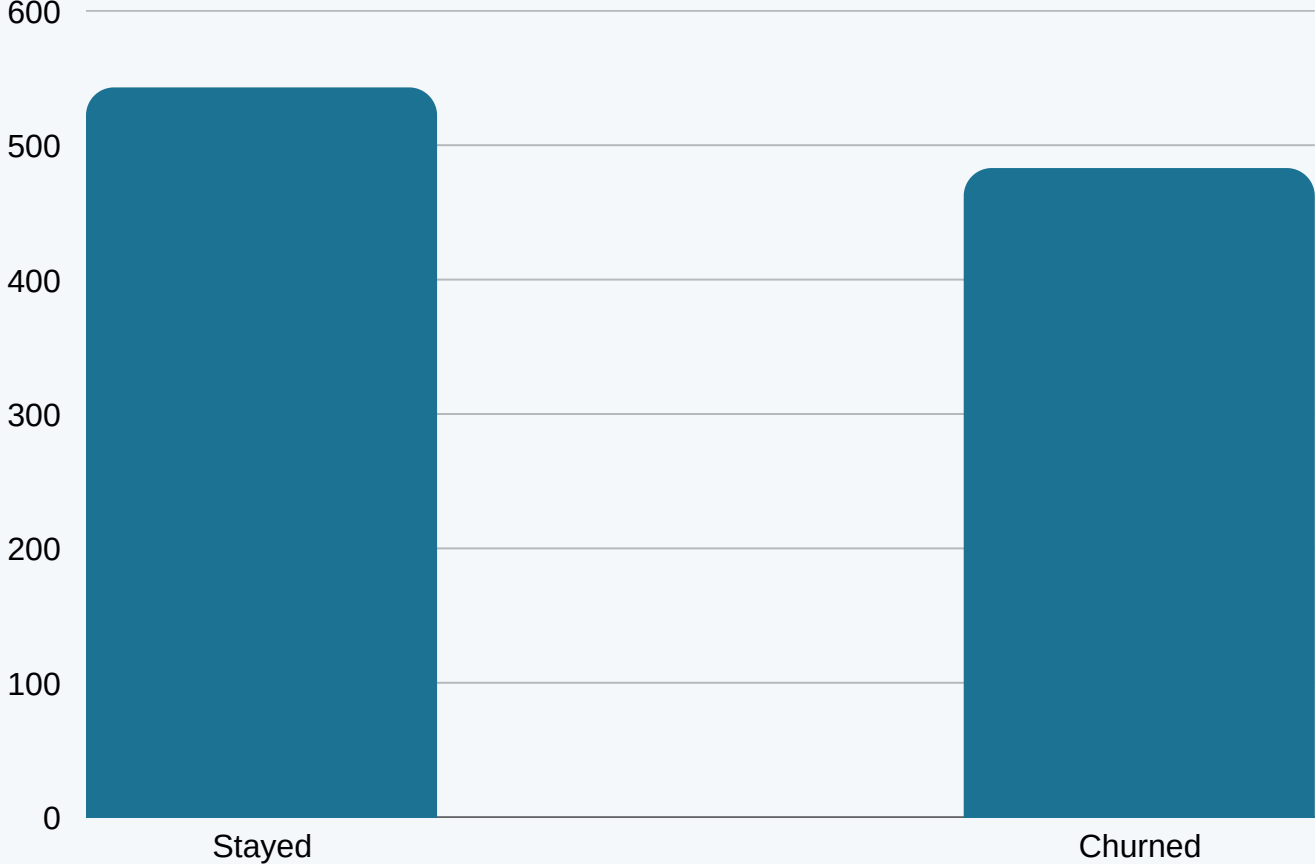
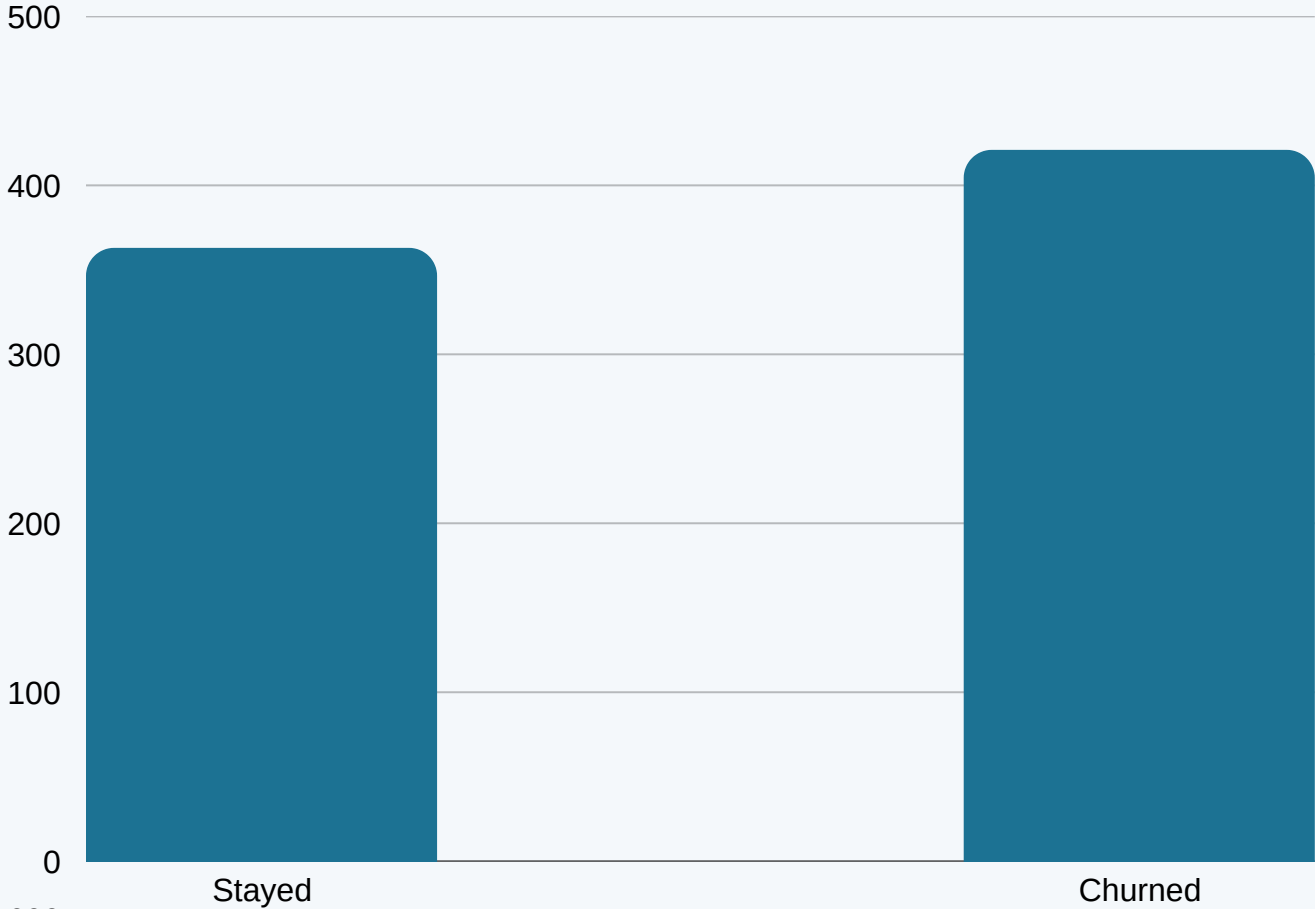
Usage decline precedes churn

Churners use 60 fewer minutes/month (483 vs 543). Low usage relative to monthly charge signals disengagement.



No leakage detected

Highest correlation with churn: 0.11 (eqpdays). No variable exceeds 0.7 — the model learns real signal.



Problem Definition — The Target Segment

Business Question: How can Company A reduce voluntary churn among its highest-value customers by addressing the root cause identified in EDA — device ageing?



Revenue filter

$\text{rev_Mean} \geq \text{Q75}$
(\$70.75/month)



Device filter

$\text{eqpdays} > 365$
(> 1 year old)



Action filter

$\text{churn_proba} \geq 0.60$
(model cutoff)

7,763

Customers:
high value + aged device

55.2%

Churn rate
in segment

49.1%

Churn rate
outside segment

\$107.19

Avg monthly
revenue

558 days

Avg device
age (~18 months)

2,798

Priority targets
($\text{churn_proba} \geq 0.60$)

Problem Definition — The Target Segment

¿Why this segment?

Although this segment represents a relatively small portion of the customer base, it generates a disproportionately large share of the company's revenue.

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(\$70.75/month)

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Problem Definition — The Target Segment



Device filter

eqpdays > 365
(> 1 year old)

55.2%

Churn rate
in segment

558 days

Avg device
age (~18 months)

¿Why this filter?

By focusing retention efforts on these high-value customers and addressing the root cause of their voluntary churn—device ageing—Company A can maximize the impact of its retention strategy.

Successfully retaining customers in this segment is expected to improve long-term revenue stability and contribute to the company's financial sustainability.

Problem Definition — The Target Segment

 Action filter

churn_proba $\geq$ 0.60
(model cutoff)

49.1%

Churn rate
outside segment

2,798

Priority targets
(churn_proba $\geq$ 0.60)

¿Why this actions filter?

A probability cutoff of 0.60 was selected because it provided the best balance among the evaluated thresholds.

At this cutoff, the model identifies 2,798 priority customers, of whom 2,207 are actual churners, yielding a precision of approximately 79%. In practical terms, this means that nearly 8 out of every 10 customers targeted by the retention strategy are true churners, reducing the number of unnecessary interventions and making the allocation of retention resources more efficient.

Model — XGBoost Classifier

Model Configuration

Algorithm	XGBoost Classifier
Training set	70,000 customers (stratified split)
Test set	30,000 customers
Estimators	300 · Learning rate: 0.05 · Max depth: 6
Random state	42 (reproducible)

Accuracy

~76%

vs 50% baseline

AUC-ROC

> 0.83

strong discrimination

Precision
(cutoff 0.60)

78.9%

*8/10 flagged are real
churners*

totmrc_Mean

change_mou

usage_per_charge

mou_Mean

hnd_price

refurb_new

months

eqpdays

0

10

20

30

40

50

Business Proposal — Proactive Device Upgrade Program

01

Identify

Model scores all 100K customers weekly. Top 2,798 high-value customers with $\text{churn_proba} \geq 0.60$ are flagged.

02

Offer

Retention team contacts flagged customers with a proactive upgrade: trade-in of old device as partial payment.

03

Finance

Remaining balance financed in 12 or 24 monthly installments. Customer stays on current plan — no friction.

04

Retain

Installment commitment creates structural retention for the duration of the financing period.

Why this lever works

- ✓ Device financing as a retention tool is standard practice among LatAm operators — Claro, Movistar, and Tigo all use installment plans to reduce voluntary churn.
- ✓ A customer with an active installment plan has a concrete financial reason to stay. The upgrade resolves the experiential driver AND creates structural lock-in simultaneously.

Why cutoff 0.60?

Cutoff	Clients	Precision
0.50	4,989	69.3%
0.60 ✓	2,798	78.9%
0.70	1,050	89.2%

Quantified Business Impact

Stated Assumptions (conservative):

2,798 clients targeted · \$120 intervention cost/client · 25% success rate · \$107.19 avg monthly revenue

Scenario: 12-Month Installment Plan	
Clients retained	700
Revenue retained	\$899,753
Program cost	\$335,760
Net impact	\$563,993
ROI	168%

Scenario: 24-Month Installment Plan	
Clients retained	700
Revenue retained	\$1,799,506
Program cost	\$335,760
Net impact	\$1,463,746
ROI	436%



Limitations & Conclusion

⚠️ Model Limitations

- Success rate (25%) is assumed, not measured — a pilot program would validate it.
- Model does not capture involuntary churn (non-payment, address change).
- No competitor pricing or market offers data — model can't see external pull factors.
- Substitution risk: retained customer may downgrade plan, reducing ARPU.
- Dataset lacks timestamps for dynamic churn — scores are point-in-time.

✓ Key Takeaways

- ✓ eqpdays is the #1 churn driver — device upgrade directly addresses the root cause.
- ✓ 2,798 priority customers identified with 78.9% model precision at cutoff 0.60.
- ✓ Net impact: \$564K (12m) to \$1.46M (24m) under conservative assumptions.
- ✓ Installment financing creates structural retention — not just a one-time offer.
- ✓ Proposal is grounded in Colombia's regulatory context (CRC, 2025) and cited industry benchmarks.

Recommendation: Launch a pilot program targeting the top 500 customers by predicted churn probability. Measure actual success rate, refine assumptions, then scale to the full 2,798 cohort.

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